# Customer Support Ticket Priority Prediction

## Project Title

Customer Support Ticket Priority Prediction and Automated Assignment System Using Agentforce

## Problem Statement

Customer support teams receive a large number of tickets daily, but prioritization and assignment are often manual. This leads to delays in resolving critical issues and poor customer satisfaction.

There is a need for a system that:

- Automatically classifies ticket priority
- Assigns tickets to the right support agent
- Reduces response time
- Uses AI to understand urgency

## User Stories

1. **As a Support Agent**, I want tickets to be automatically prioritized so that I can focus on critical issues first.
2. **As a Customer**, I want urgent issues to be resolved quickly so that my experience improves.
3. **As a Manager**, I want tickets to be assigned automatically to the right team so that workload is balanced.
4. **As an AI Agent (Agentforce)**, I want to analyze ticket details and trigger automation so that manual work is reduced.

## Expected Outcome

- Faster ticket resolution

- Improved customer satisfaction
- Reduced manual effort
- Intelligent ticket handling
- Better team productivity

# Custom Object

## Object Name:

Support Ticket Intelligence

## API Name:

`Support_Ticket_Intelligence__c`

# Fields

| Field Label | API Name | Data Type | Description |
|---|---|---|---|
| Ticket Number | Ticket_Number__c | Auto Number | TKT-{0000} |
| Customer | Customer__c | Lookup (Account) | Related Account |
| Contact | Contact__c | Lookup (Contact) | Customer contact |
| Issue Type | Issue_Type__c | Picklist | Technical, Billing, General |
| Description | Description__c | Long Text Area | Issue details |
| Priority Level | Priority_Level__c | Picklist | Low, Medium, High |
| Status | Status__c | Picklist | New, In Progress, Resolved |
| Created Date | Created_Date__c | Date | Ticket date |
| Assigned To | Assigned_To__c | Lookup (User) | Support agent |
| SLA Breach Risk | SLA_Breach_Risk__c | Checkbox | Risk flag |

| | | | |
|---|---|---|---|
| Resolution Time (hrs) | Resolution_Time__c | Number | Time taken |

# Auto-Launched Flow Logic (Modified for this Project)

# 1. Flow Type

Auto-Launched Flow (No Trigger or called from Agentforce)

# 2. Variables

## Input Variable

| Label | API Name | Type | Input |
|---|---|---|---|
| Account Name | `varAccountName` | Text | Yes |

## Output Variables

| Label | API Name | Type | Output |
|---|---|---|---|
| Account Id | `varAccountId` | Text | Yes |
| Ticket Id | `varTicketId` | Text | Yes |

| | | | |
|---|---|---|---|
| Priority Level | `varPriorityLevel` | Text | Yes |
| Assigned Agent | `varAssignedTo` | Text | Yes |
| Action Message | `varActionMessage` | Text | Yes |

# 3. Step-by-Step Flow Implementation

## Step 1: Get Account

**Element:** Get Records

- Label: Get Account
- Object: Account
- Condition:
  - Name Equals `{!varAccountName}`
- Sort: CreatedDate DESC
- Store: Only first record

## Step 2: Assignment (Store Account Id)

**Element:** Assignment

- Label: Store Account Id

Set:

- `varAccountId = {!Get_Account.Id}`

## Step 3: Get Ticket Record

**Element:** Get Records

- Label: Get Ticket
- Object: `Support_Ticket_Intelligence__c`
- Condition:
  - Customer__c Equals `{!varAccountId}`
- Sort: CreatedDate DESC
- Store: Only first record

# Step 4: Assignment (Store Ticket Id)

**Element:** Assignment

- Label: Store Ticket Id

Set:

- `varTicketId = {!Get_Ticket.Id}`

# Step 5: Decision (Analyze Description)

**Element:** Decision

- Label: Analyze Description

## Outcomes:

### High Priority

- Description contains:
  - "urgent"
  - "not working"
  - "failure"

(Use multiple conditions with CONTAINS)

### Medium Priority

- Description contains:
  - "issue"
  - "slow"
  - "delay"

### Default Outcome

- Low Priority

# Step 6: Assignment (Set Priority Level)

**High Path**

- `varPriorityLevel = "High"`

**Medium Path**

- `varPriorityLevel = "Medium"`

**Low Path**

- `varPriorityLevel = "Low"`

# Step 7: Decision (High Priority Check)

**Element:** Decision

- Label: Is High Priority

Condition:

- `varPriorityLevel Equals "High"`

# Step 8: Create Task (Only High Priority)

**Element:** Create Records

- Object: Task

**Set Fields:**

- Subject = "Urgent Ticket Handling"
- WhatId = `{!varTicketId}`
- Priority = High
- Status = Not Started

# Step 9: Assignment (Assign Agent Name)

**Element:** Assignment

- Label: Set Assigned Agent

**Set**:

- `varAssignedTo = "Senior Support Agent"`

## Step 10: Optional SLA Check

**Decision:** SLA Risk

Condition example:

- If Created Date older than 2 days

Then:

- (Optional internal flag or message)

## Step 11: Final Output Message

**Element:** Assignment

- Label: Set Final Message

Set `varActionMessage`:

- High:
  "High priority ticket detected. Assigned to senior agent."
- Medium:
  "Ticket marked as medium priority. Will be handled shortly."
- Low:
  "Tickets are low priority and queued for processing."

## Step 12 : Save Flow

Label Name : Support_Ticket_Intellegence

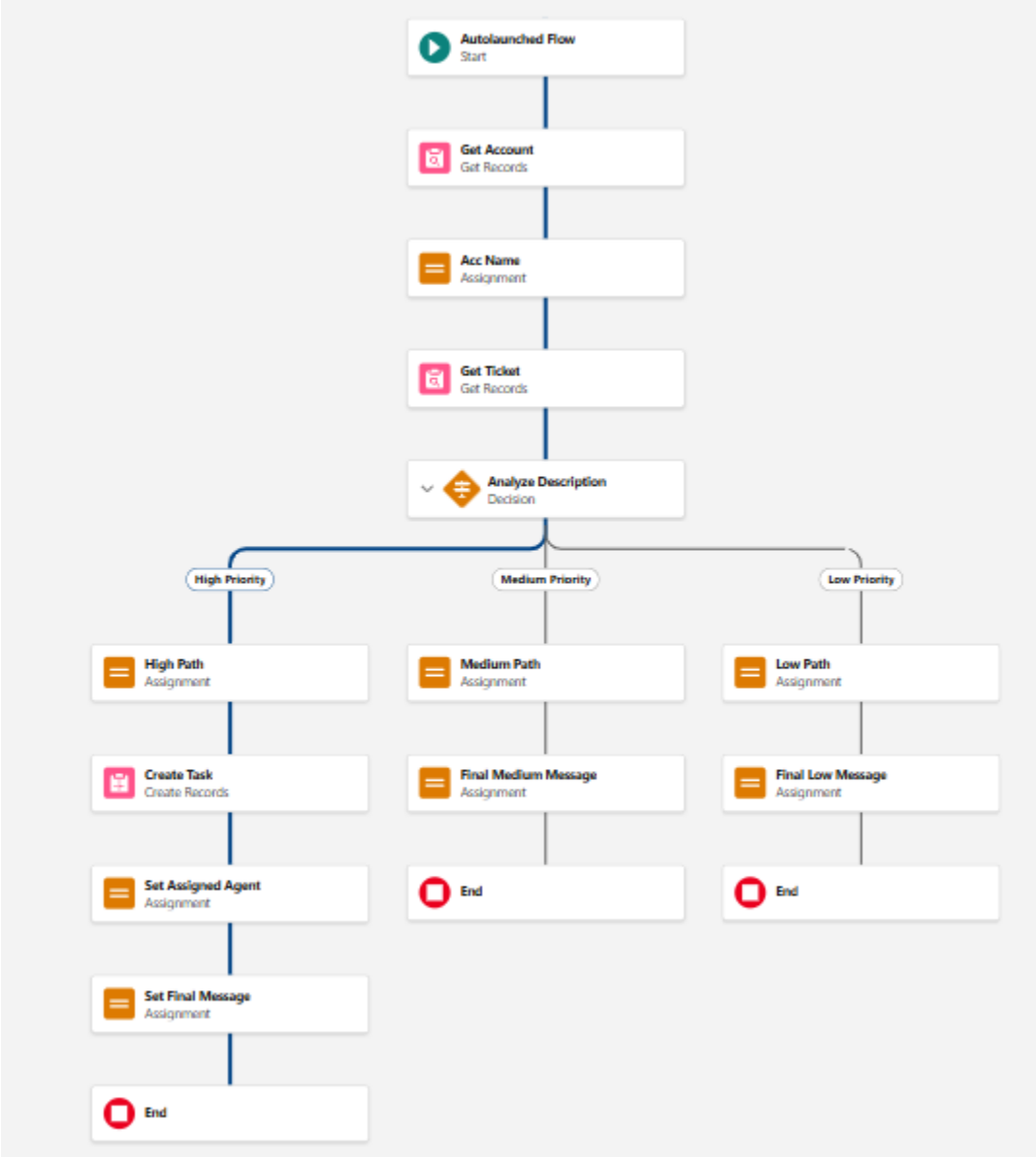
Active Flow

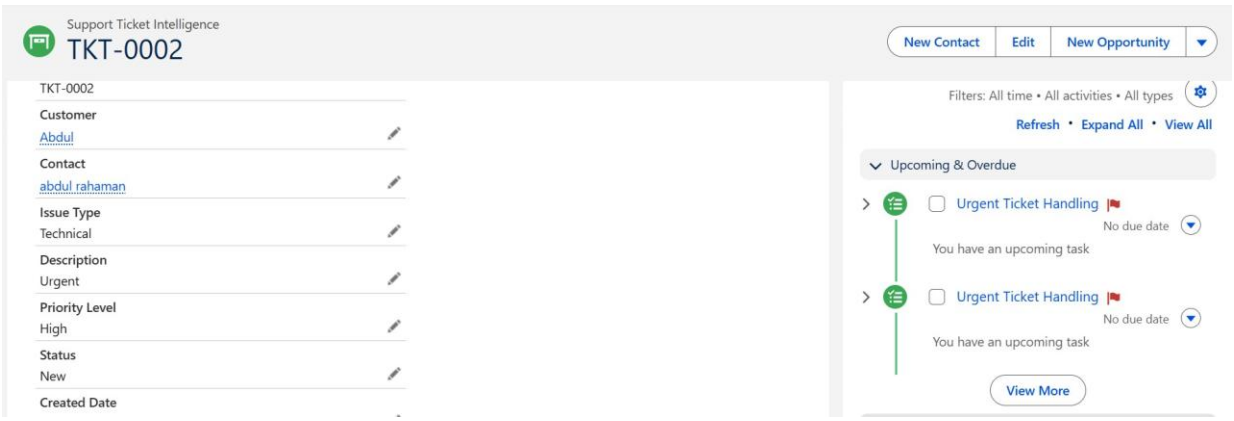
# 4. Final Flow Structure

Start
→ Get Account
→ Assignment (Account Id)
→ Get Ticket

→ Assignment (Ticket Id)
→ Decision (Analyze Description)
→ Assignment (Priority Level)
→ Decision (High Priority)
→ Create Task
→ Assignment (Assigned Agent)
→ Optional SLA Decision
→ Assignment (Final Message)
→ End

**Flow OutPut :**





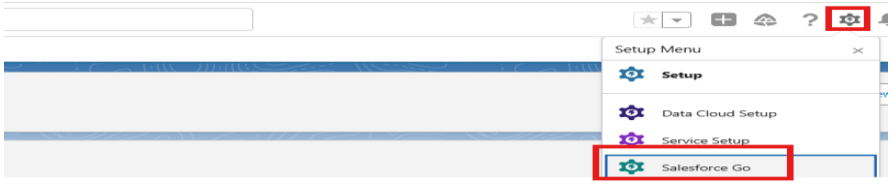

# 5. Final Outcome

- Takes **Account Name as input**
- Fetches latest support ticket
- Analyzes description using keywords
- Sets priority automatically
- Creates task for urgent cases
- Returns output to Agentforce

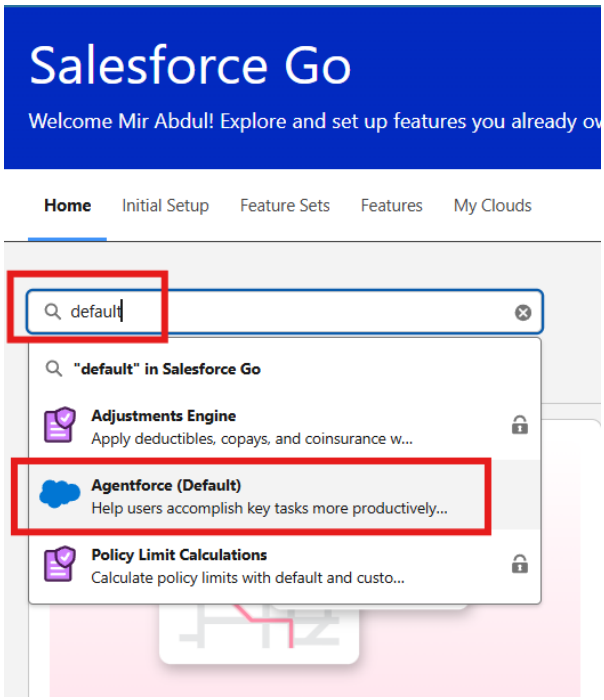
## Agentforce Configuration

Steps :

1. Go to Gear icon and Click Salesforce Go

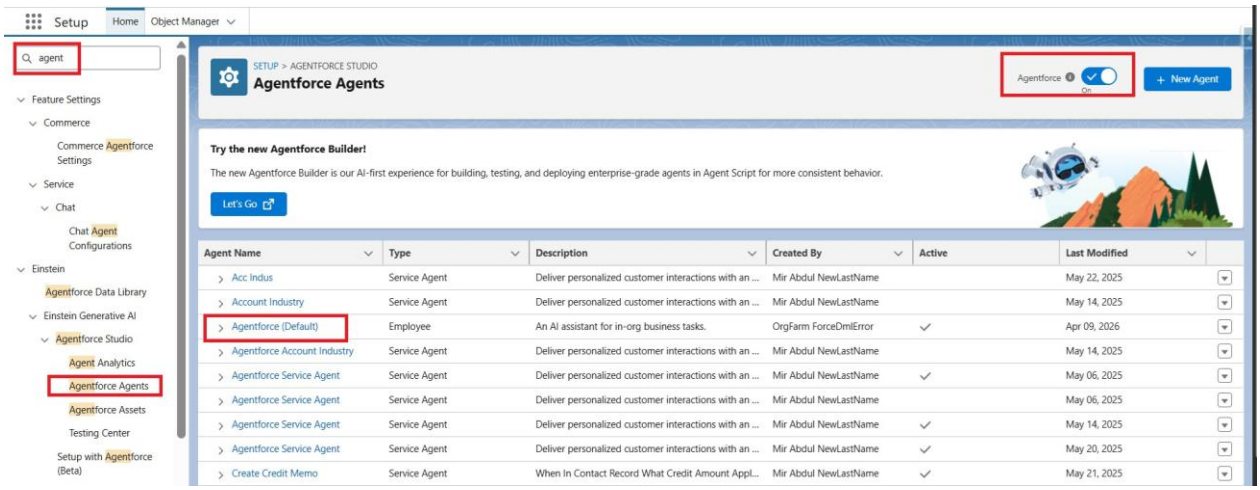



2. Search Default and Select Agentforce Default then Enabled It

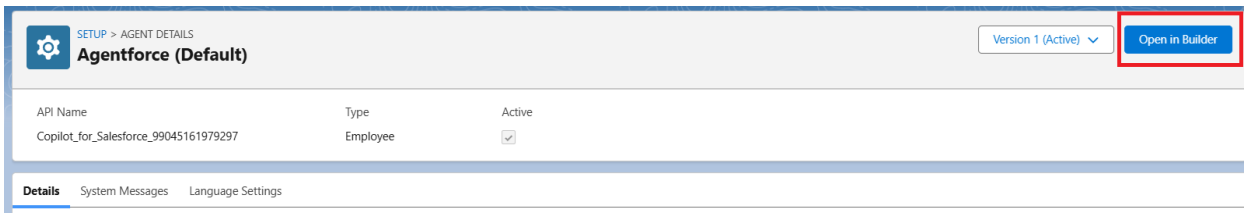



3. Select Get Started ⇒ Turn On ⇒ Confirm

4. Search Agents in Quick Find Box and Enabled The Agent then you will see Default Agent

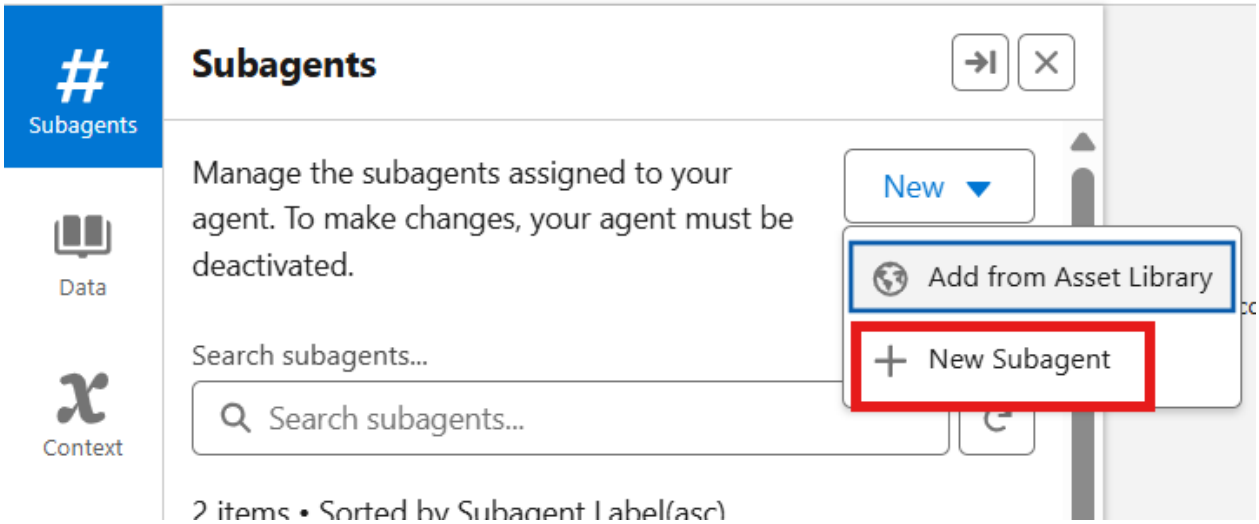



5. Select Default Agent and Click Open in Builder

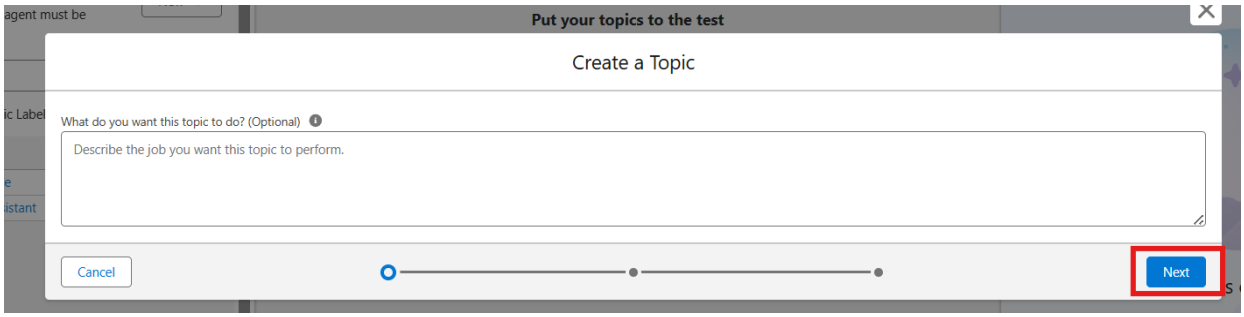


6. Click New to Create New SubAgent




7. Click Next




## Agentforce Subagent Configuration

# Name

Support Ticket Priority Analysis

# API Name

Support_Ticket_Priority_Analysis

# Classification Description

This topic analyzes customer support tickets to determine priority based on issue description and urgency. It handles user requests related to ticket status, issue severity, and support prioritization.

# Scope

Your job is only to analyze support ticket descriptions, determine the priority level (High, Medium, Low), and trigger backend automation for task assignment. Do not handle unrelated requests such as billing, subscription management, or account updates.

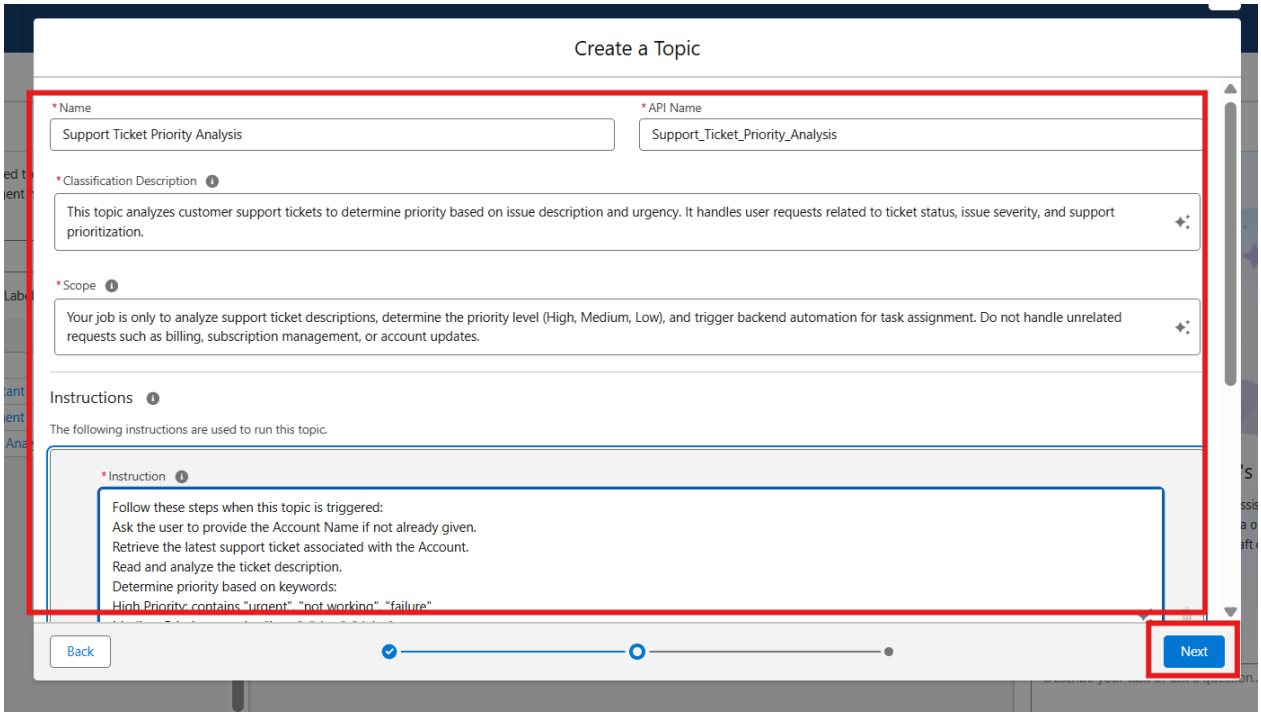
# Instructions

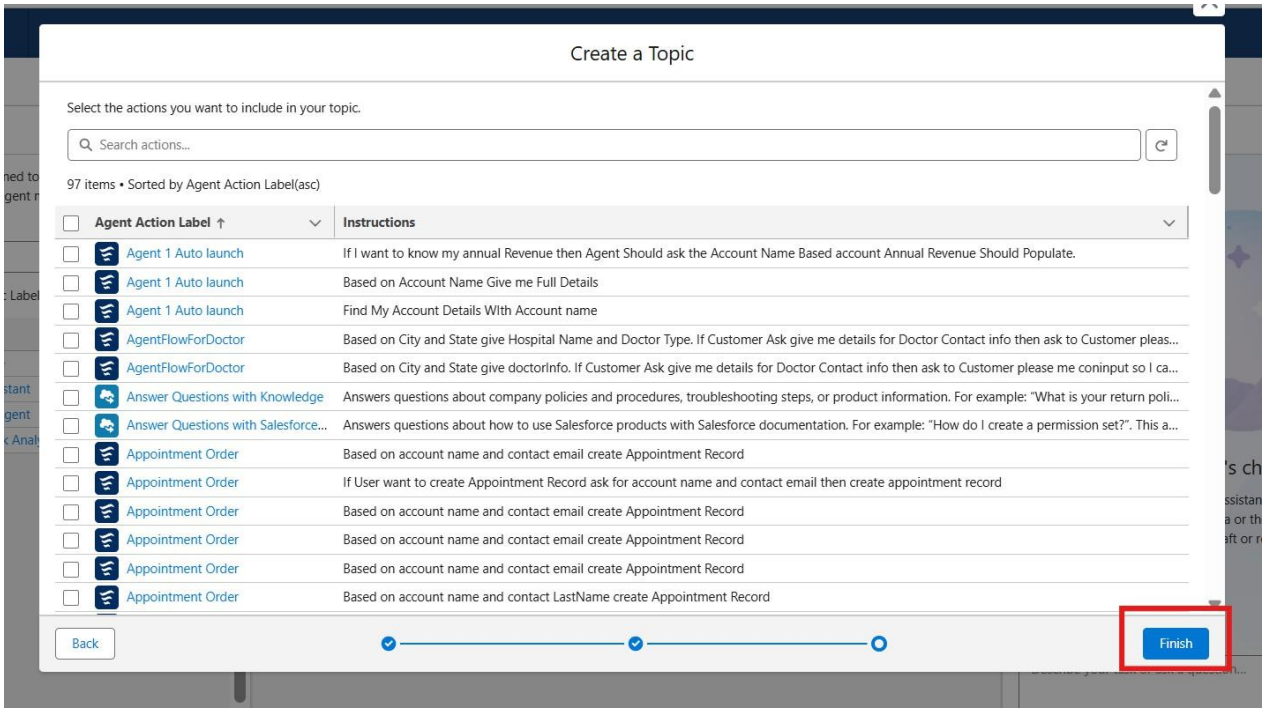
Follow these steps when this topic is triggered:

1. Ask the user to provide the Account Name if not already given.
2. Retrieve the latest support ticket associated with the Account.
3. Read and analyze the ticket description.
4. Determine priority based on keywords:
    - High Priority: contains "urgent", "not working", "failure"
    - Medium Priority: contains "issue", "slow", "delay"
    - Low Priority: if none of the above
5. Set the priority level accordingly.
6. If the ticket is High Priority:
    - Trigger the Auto-Launched Flow
    - Ensure a task is created for immediate handling
7. Assign the ticket to the appropriate support level (e.g., senior agent for high priority).
8. Return a clear response message to the user:
    - High → Inform urgent handling is initiated
    - Medium → Inform issue will be handled shortly

- Low → Inform ticket is queued

9. Do not update records directly or send emails. Use Flow for all actions.




Select Topic

SMARTBRIDGE

salesforce PARTNER

TJUlii.bQa.wi WORKFORCE PARTNER

## Topics

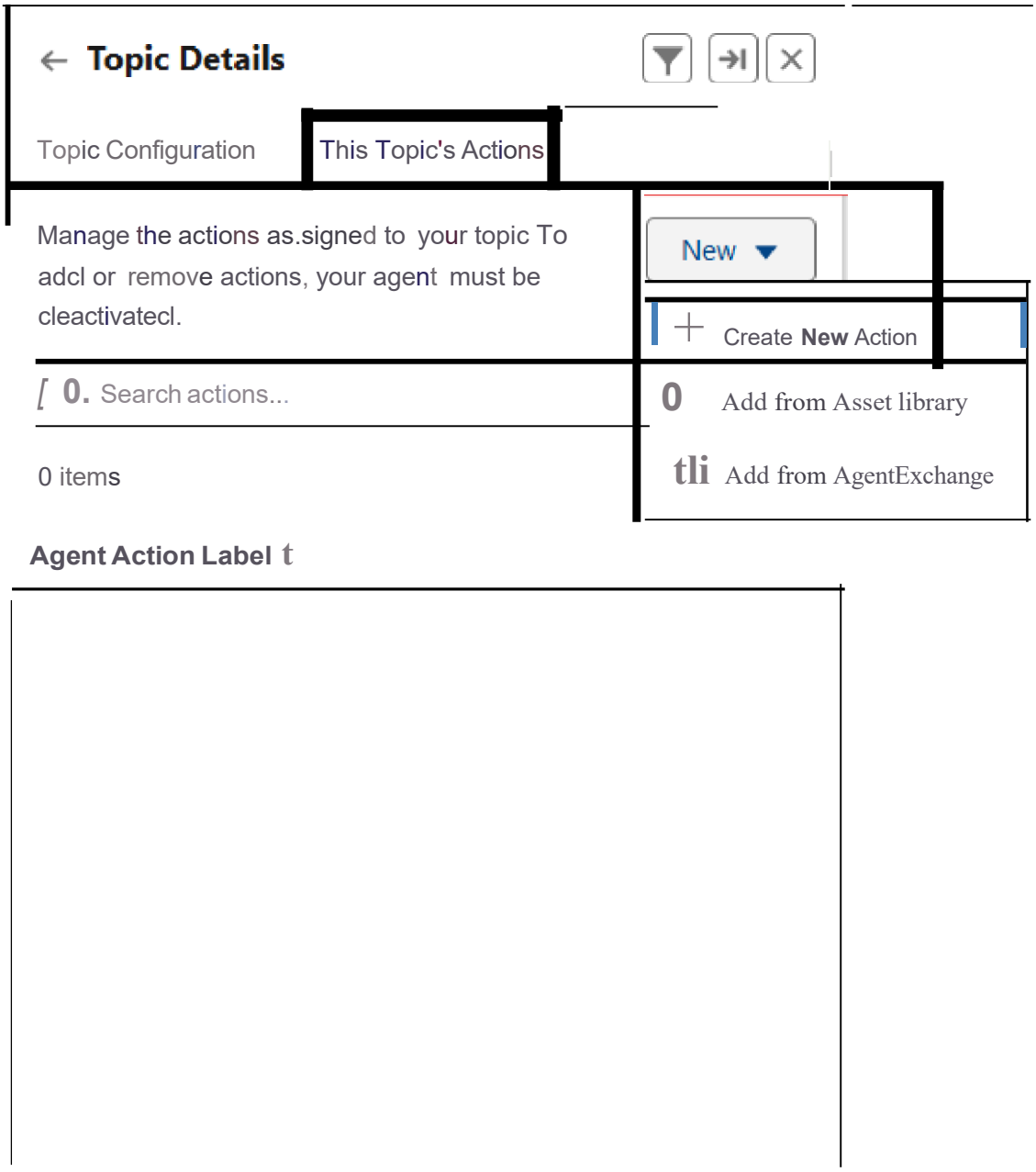
Manage the topics assigned to your age11t. To make changes, your agent must be deactivated.

New.,..

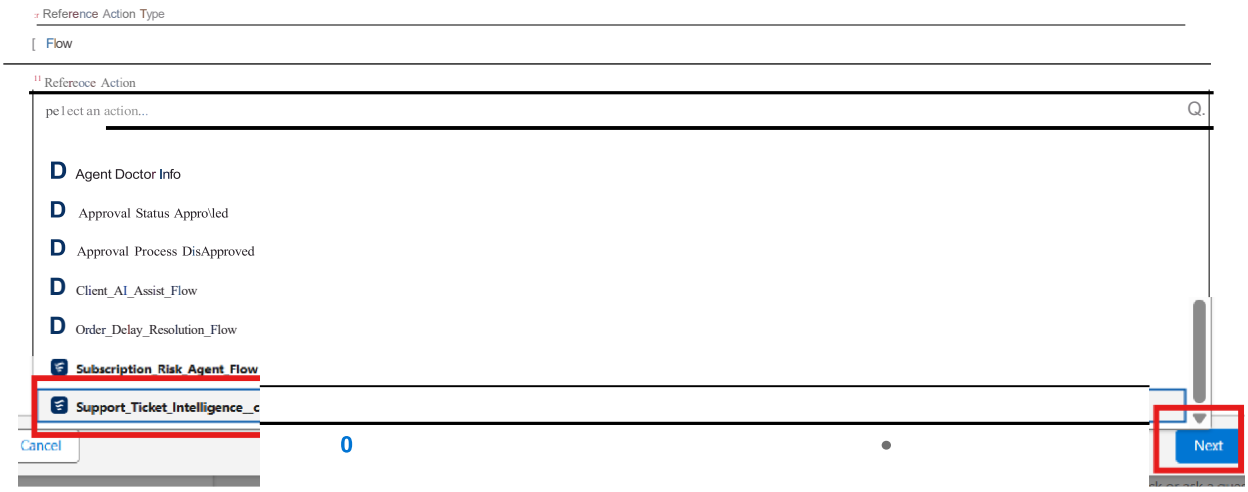
_[ o. se_a_r_ch_to_p_i_cs•_•_.

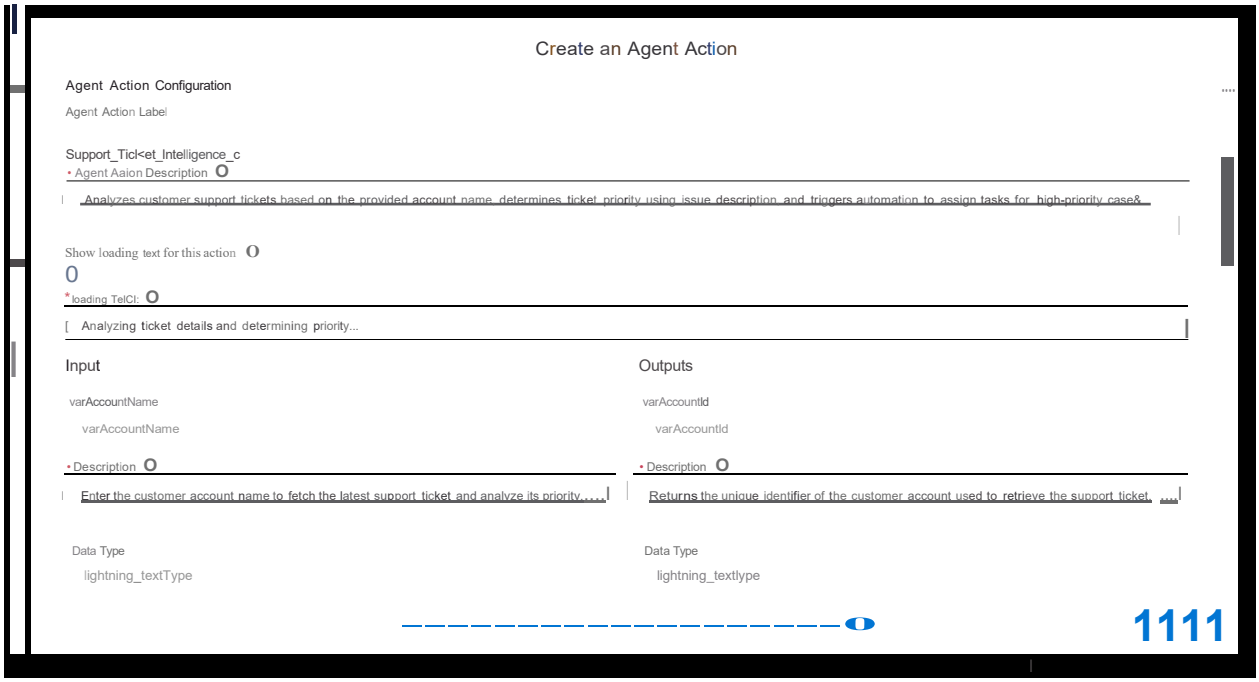
5 items • Sorted by Topic Label(asc)

| Topic Label t |
| --- |
| Client Support Assistance |
| Construction Project Assistant |
| Order Delay Resolution Agent |
| Subscription Renewal Risk Analysis |
| SUQQOrt Ticket Priority Alli!!Y..ill |


Topic Details
Topic Configuration
This Topic's Actions
Manage the actions as.signed to your topic To adcl or remove actions, your agent must be cleactivatecl.
New
Create New Action
[ 0. Search actions...
0 Add from Asset library
0 items
tli Add from AgentExchange
Agent Action Label t

Actions are how an agent gets things done. To create an agent action, start with the functionality you already have In Salesforce, such as flom or prompt templates. Select an action you want the agent action to reference, and then configure it for use with an agent.



# Agent Action Description

Analyzes customer support tickets based on the provided account name, determines ticket priority using issue description, and triggers automation to assign tasks for high-priority cases.

# Loading Text

Analyzing ticket details and determining priority...

# Input

## varAccountName

- **Description**
  Enter the customer account name to fetch the latest support ticket and analyze its priority.
- **Data Type**
  lightning__textType
- **Require Input**
  Enabled
- **Collect Data From User**
  Enabled

# Outputs

## varAccountId

- **Description**
  Returns the unique identifier of the customer account used to retrieve the support ticket.
- **Data Type**
  lightning__textType
- **Show in Conversation**
  Enabled
- **Output Rendering**
  Text

## varActionMessage

- **Description**
  Displays the final message indicating ticket priority and actions taken by the system.
- **Data Type**
  lightning__textType
- **Show in Conversation**
  Enabled
- **Output Rendering**
  Text

## varAssignedTo

- **Description**
  Indicates the support agent or team assigned to handle the ticket.
- **Data Type**
  lightning__textType
- **Show in Conversation**
  Enabled
- **Output Rendering**
  Text

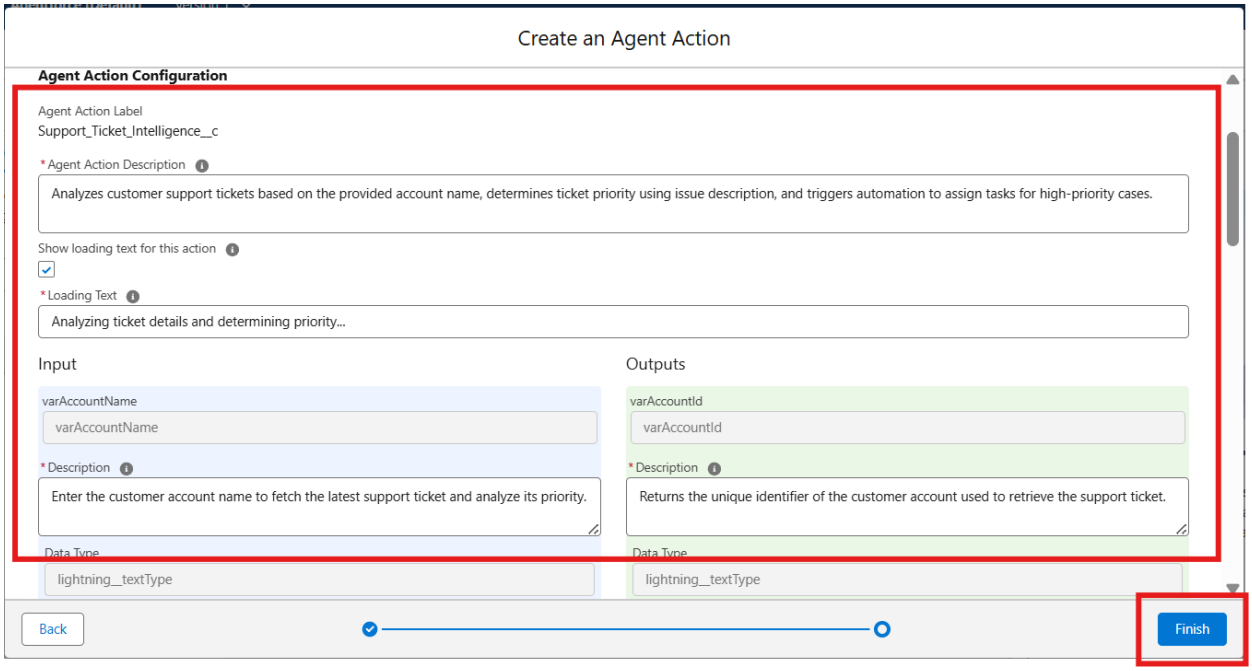
## varPriorityLevel

- **Description**
  Represents the calculated ticket priority level (High, Medium, Low) based on issue description.
- **Data Type**
  lightning__textType
- **Show in Conversation**
  Enabled
- **Output Rendering**
  Text

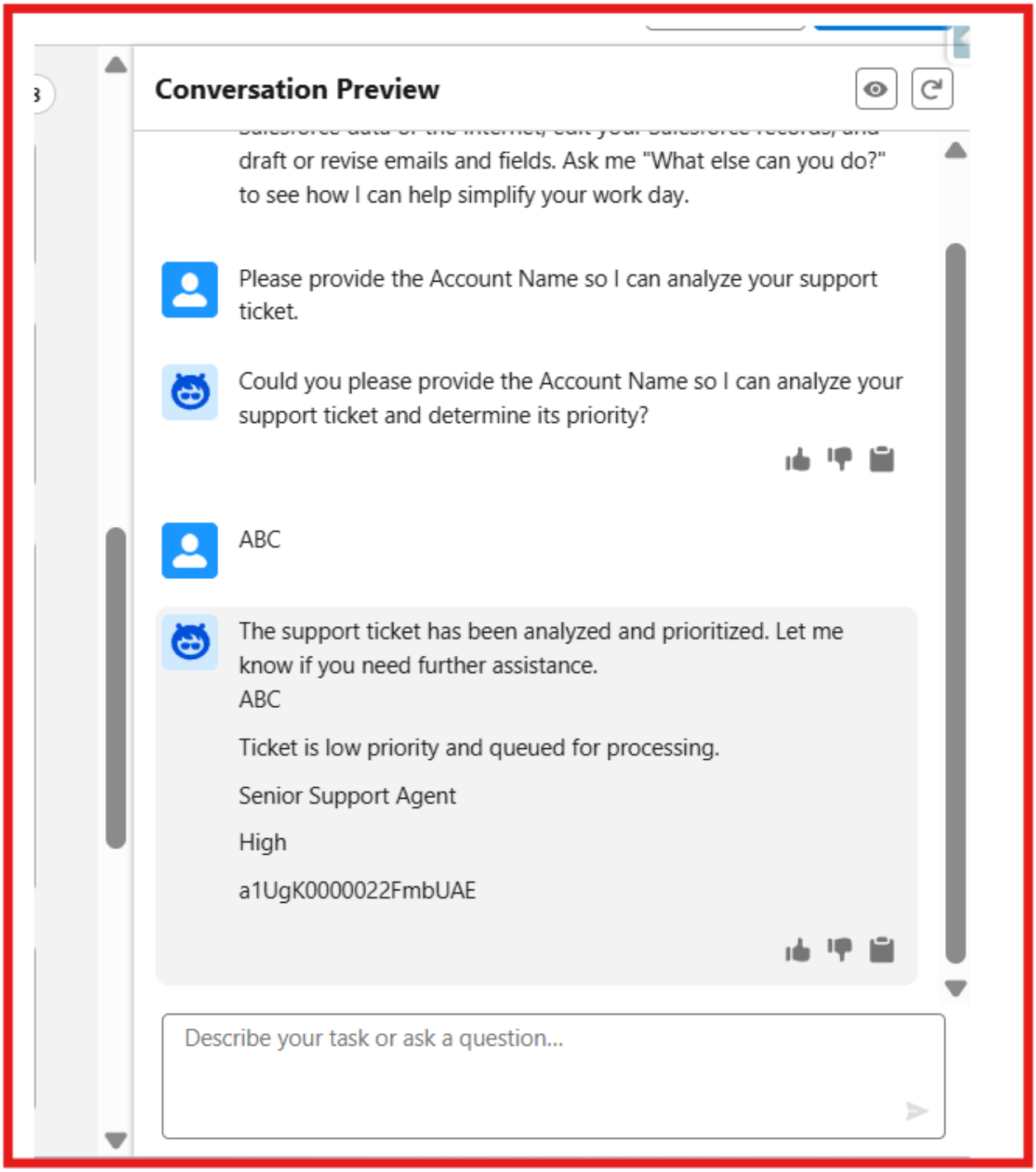
## varTicketId

- **Description**
  Returns the unique identifier of the support ticket analyzed by the system.
- **Data Type**
  lightning__textType
- **Show in Conversation**
  Enabled

- **Output Rendering**
  Text




**Final OutPut :**




**Note: If you would like to extend this scenario, you can add more variables to the flow and configure them in the agent. This will help you achieve better outputs. It's a great way to learn. If you'd like to explore more scenarios, feel free to get in touch with me.**

**Github link:**

https://github.com/editorzplayer-cyber/Customer-Support-Ticket-Priority-Prediction-.git

**Demo Video Link:**

https://drive.google.com/file/d/14hoEpkIJsOe9pDxTU8kfS6bPPAEjRs9/view?usp=sharing